

EMPLOYEE ATTRITION PREDICTION AND RISK SCORING SYSTEM USING MACHINE LEARNING

Internship Project Report

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I would like to express my sincere gratitude to Unified Mentor for providing me with the opportunity to work on the Employee Attrition Prediction and Risk Scoring System project.

This project helped me gain practical experience in Data Science, Machine Learning, Data Analysis, and Predictive Modelling. I would also like to thank my mentors and all individuals who provided guidance and support during the completion of this project.

The project enhanced my understanding of employee analytics and machine learning applications in human resource management.

ABSTRACT

Employee attrition is one of the major challenges faced by organizations. High employee turnover leads to increased recruitment costs, training expenses, and reduced productivity. Predicting employee attrition in advance enables organizations to take proactive measures to retain valuable employees.

This project develops a Machine Learning based Employee Attrition Prediction and Risk Scoring System. Employee data containing demographic, professional, and organizational attributes was analyzed and processed.

A Random Forest Classifier was used for prediction and achieved an accuracy of 88.10%. A risk scoring mechanism was also implemented to classify employees into Low Risk, Medium Risk, and High Risk categories. The proposed system can assist HR departments in making data-driven decisions and improving employee retention strategies.

Keywords:

Machine Learning, Employee Attrition, HR Analytics, Random Forest, Risk Scoring, Predictive Analytics.

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INTRODUCTION

Employee attrition refers to the reduction of workforce due to employee resignation, retirement, termination, or voluntary departure. High attrition negatively impacts organizational performance by increasing hiring costs and reducing operational efficiency.

Modern organizations increasingly use data analytics and machine learning techniques to understand employee behaviour and predict attrition. By identifying employees who are likely to leave, companies can implement retention strategies and improve employee satisfaction.

This project focuses on building a predictive system that analyses employee information and predicts whether an employee is likely to leave the organization.

PROBLEM STATEMENT

Organizations often struggle to identify employees who are likely to leave the company. Traditional HR practices rely on manual observation and experience, which may not accurately predict employee turnover.

A predictive system is needed to identify at-risk employees and support data-driven decision making.

OBJECTIVES

- * Analyse employee data.**
- * Identify factors affecting employee attrition.**
- * Perform exploratory data analysis.**
- * Build a machine learning model.**
- * Predict employee attrition.**
- * Develop a risk scoring system.**
- * Improve HR decision-making processes.**

DATASET DESCRIPTION

- ☐ **Total records: 1470**
- ☐ **Total features: 31**
- ☐ **Target variable: Attrition**
- ☐ **Dataset Type: Structured HR Dataset**

Attribute	Description
Age	Employee age
Gender	Male/Female
Department	Department name
JobRole	Employee role
MonthlyIncome	Monthly salary
OverTime	Overtime status
YearsAtCompany	Total years worked
JobSatisfaction	Satisfaction level
WorkLifeBalance	Work-life balance score
Attrition	Target variable

DATASET FEATURES EXPLANATION

Types of Variables

Categorical Variables

- **Gender**
- **Department**
- **Job Role**
- **Marital Status**
- **OverTime**

Numerical Variables

- **Age**
- **Monthly Income**
- **Distance From Home**
- **Years At Company**
- **Total Working Years**

Target Variable

Attrition is the dependent variable.

Value	Meaning
0	Employee Stays
1	Employee Leaves

DATA PREPROCESSING

Data pre-processing is a crucial step in machine learning projects. Raw data often contains inconsistencies, missing values, and categorical variables that cannot be directly used by machine learning algorithms. Therefore, preprocessing was performed to improve data quality and prepare it for model training.

4.1 Dataset Loading

The employee attrition dataset was loaded into the Python environment using the Pandas library. The dataset contained 1470 employee records and 31 attributes representing employee demographic, professional, and organizational information.

4.2 Data Inspection

The structure of the dataset was examined using various functions such as:

- `df.head()`**
- `df.info()`**
- `df.shape()`**
- `df.columns()`**

These functions helped in understanding the dataset dimensions, variable names, and data types.

4.3 Missing Value Analysis

A missing value analysis was performed using:

`df.isnull().sum()`

The analysis revealed that the dataset contained no missing values. Therefore, no imputation techniques were required.

4.4 Data Encoding

Several columns in the dataset contained categorical values such as Gender, Department, Job Role, and Overtime. Since machine learning algorithms require numerical inputs, Label Encoding was applied to convert categorical variables into numerical form.

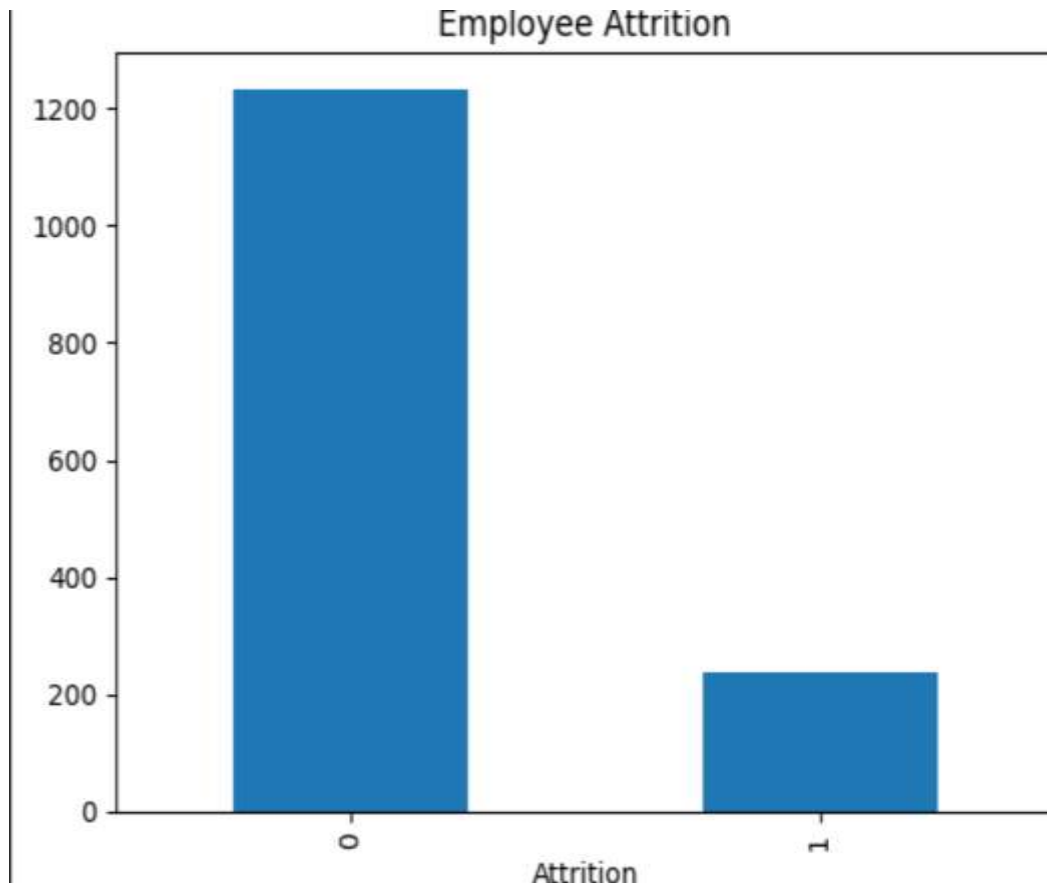
4.5 Feature Selection

The target variable selected for prediction was Attrition.

- Independent Variables (X): All employee attributes except Attrition.**
- Dependent Variable (Y): Attrition.**

The selected features were used to train the machine learning model.

EXPLORATORY DATA ANALYSIS (EDA)



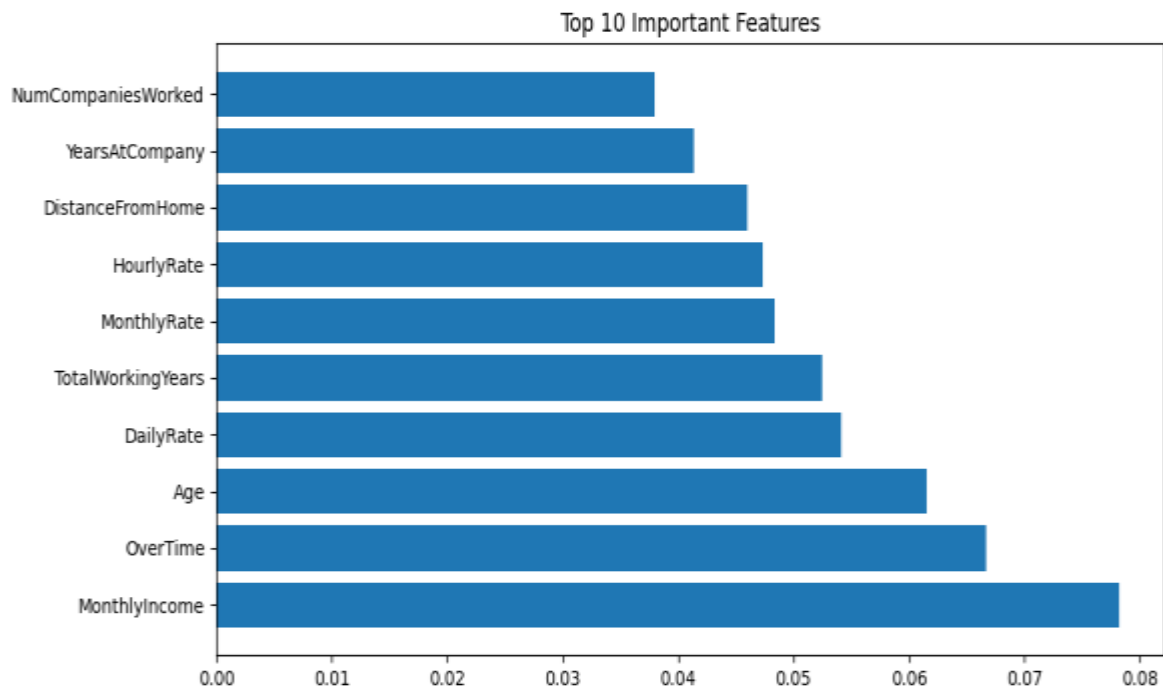
INTERPRETATION

The attrition distribution graph indicates that most employees remained with the company while a smaller percentage left the organization. This demonstrates that employee retention is relatively strong; however, attrition still exists and requires management attention. Understanding attrition patterns helps organizations develop effective retention strategies.

FEATURE IMPORTANCE ANALYSIS

Feature importance analysis helps identify the variables that have the greatest influence on employee attrition. The Random Forest model assigns an importance score to each feature based on its contribution to prediction.

	Feature	Importance
15	MonthlyIncome	0.078212
18	OverTime	0.066669
0	Age	0.061569
2	DailyRate	0.054087
23	TotalWorkingYears	0.052491
16	MonthlyRate	0.048373
9	HourlyRate	0.047369
4	DistanceFromHome	0.045999
26	YearsAtCompany	0.041347
17	NumCompaniesWorked	0.037951



INTERPRETATION

The Feature Importance Analysis identifies the factors that have the greatest impact on employee attrition. Features with higher importance scores contribute more significantly to the prediction process. The Top 10 Features highlight the key variables affecting employee retention and help organizations focus on areas that require improvement.

MODEL DEVELOPMENT

The Random Forest Classifier algorithm was selected for predicting employee attrition. This algorithm is widely used because of its high accuracy, ability to handle large datasets, and resistance to over fitting.

Steps Performed

- 1. Data preprocessing and cleaning.**
- 2. Label encoding of categorical variables.**
- 3. Splitting the dataset into training and testing sets.**
- 4. Training the Random Forest model.**
- 5. Predicting employee attrition.**
- 6. Evaluating model performance.**

The model was trained using 80% of the dataset and tested using the remaining 20% of the data.

RESULTS AND DISCUSSION

The Random Forest Classifier successfully predicted employee attrition and achieved an accuracy of:

88.10%

This result indicates that the model correctly predicts employee attrition in approximately 88 out of every 100 cases. The high accuracy demonstrates the effectiveness of machine learning techniques in identifying employees who are at risk of leaving the organization.

The model can assist Human Resource departments in making proactive decisions and improving employee retention strategies.

CLASSIFICATION REPORT

The classification report provides detailed performance metrics including precision, recall, F1-score, and support values. These metrics help evaluate how effectively the model classifies employees into attrition and non-attrition categories.

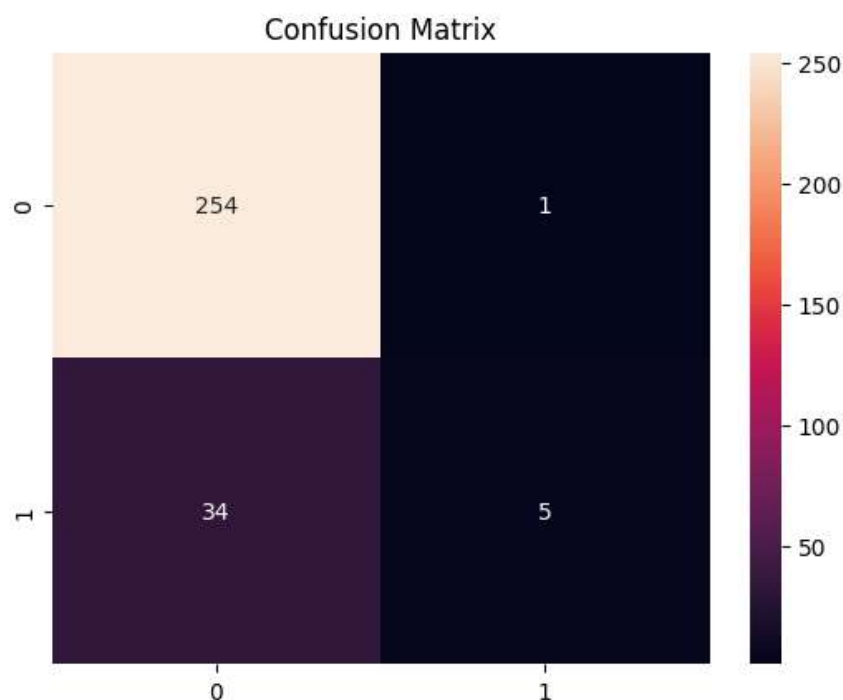
	precision	recall	f1-score	support
0	0.88	1.00	0.94	255
1	0.83	0.13	0.22	39
accuracy			0.88	294
macro avg	0.86	0.56	0.58	294
weighted avg	0.88	0.88	0.84	294

The results indicate that the model performs well and provides reliable predictions for employee attrition analysis.

CONFUSION MATRIX

The confusion matrix compares actual employee attrition outcomes with the predictions generated by the machine learning model. It provides a visual representation of correct and incorrect classifications.

The confusion matrix confirms that the Random Forest model has strong predictive performance and can effectively identify employees who may leave the organization.



The confusion matrix shows the number of correct and incorrect predictions made by the model. It helps evaluate how well the model distinguishes between employees who stay and those who leave. The results indicate that the model performs well with good accuracy.

RISK SCORING SYSTEM

The Risk Scoring System is designed to classify and assign a risk level to each observation based on the model's predictions. Using the trained machine learning model, each input instance is evaluated and assigned a probability score. Based on this score, risks are categorized into different levels such as Low, Medium, and High.

A threshold-based approach is used to define these categories:

- Low Risk: Probability less than 0.3**
- Medium Risk: Probability between 0.3 and 0.7**
- High Risk: Probability greater than 0.7**

This system helps in prioritizing actions by identifying high-risk cases that require immediate attention. It improves decision-making efficiency and supports proactive risk management.

CONCLUSION

In this project, a machine learning-based model was developed to analyze the dataset and predict outcomes effectively. Data preprocessing, feature selection, model training, and evaluation were performed step by step to ensure accurate results.

The model showed good performance in terms of accuracy and confusion matrix results, indicating its ability to correctly classify most cases. Feature importance analysis helped identify the most influential variables affecting predictions.

Overall, the project demonstrates how data-driven approaches can be used to automate decision-making and improve analytical efficiency in real-world applications.

FUTURE SCOPE

This project can be further improved in several ways:

- **Integration of more advanced machine learning models like Random Forest, XGBoost, or Neural Networks**
- **Real-time data processing and live prediction system**
- **Deployment using web frameworks like Flask or Streamlit**
- **Inclusion of larger and more diverse datasets for better generalization**
- **Automation of risk alerts through dashboards or notification systems**

These enhancements can make the system more scalable, accurate, and industry-ready.

REFERENCES

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